# Indian Institute of Technology (IIT) Roorkee

**AI Prompt Usage Log – Cybersecurity Project**

**Project Title:** Data Security Risk Assessment and Mitigation using Generative AI
**Reporting Period:** 1 February 2026 – 14 February 2026
**Department:** Cybersecurity & Information Assurance
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## 1. Introduction

This document records how AI tools were used during the IIT Roorkee cybersecurity project on data security risk assessment and mitigation using Generative AI. It mirrors the daily reporting format of the main project report, with entries for each day in the 1–14 February 2026 period.

Each day describes the activities performed with AI tools, the key findings or insights, how Generative AI contributed, and the overall outcome for the project work.

## 2. Daily AI Usage Log

### Day 1 - 01 February 2026

**Activities Performed:**
Used ChatGPT in the role of an assessment coordinator to interpret the IIT project requirements and turn them into a structured evaluation checklist for the threat intelligence report.

**Key Findings:**
The assignment guidelines contained many sections and implicit expectations that were easy to miss when reading them informally.

The AI-generated breakdown clarified what was expected in the Executive Summary, Threat Overview, Attack Lifecycle, Detection Methods, Mitigation, and AI Reflection sections.

**Generative AI Contribution:**
ChatGPT produced a detailed checklist with concrete verification points for each section, such as whether the executive summary included clear recommendations and whether the threat overview contained real case studies and prioritized controls.

**Outcome:**
The checklist became the primary tool for tracking completeness and quality across the report. It was cross-checked against the official IIT guidelines to confirm alignment before being used for internal reviews.

## Day 2 - 02 February 2026

**Activities Performed:**
Consulted Perplexity to understand what makes a cybersecurity threat analysis comprehensive and credible in academic and professional contexts.

**Key Findings:**
A strong threat analysis requires clear definitions, real-world examples with data, explanation of technical processes, sector-specific impact discussion, and proper citations. It must demonstrate both breadth and depth instead of simply listing disconnected facts.

**Generative AI Contribution:**
Perplexity summarized best practices from recognised threat intelligence reports and translated them into practical criteria that could be applied directly to the team's threat overview section.

**Outcome:**
These criteria were used to review and refine the threat overview, identifying where more depth, examples, or explanation were needed. The revised section was then compared with industry reports (such as IBM and Verizon) to validate its quality level.

## Day 3 - 03 February 2026

**Activities Performed:**
Used Claude to identify common formatting and structural issues in technical cybersecurity reports and to understand how they might affect evaluation.

**Key Findings:**
Inconsistent heading styles, missing transitions between sections, citation format mismatches, and poorly formatted tables are frequent problems. Visual elements also lose impact when they lack captions or are not referenced in the text.

**Generative AI Contribution:**
Claude listed specific issues to look for and suggested consistent approaches for headings, transitions, citation formats, and table/figure presentation.

**Outcome:**
Based on this guidance, a formatting guide was created for the team. The entire document was reviewed and updated to follow the same structure and style, improving readability and presentation consistency.

## Day 4 - 04 February 2026

**Activities Performed:**
Asked ChatGPT how to balance technical depth with readability for an academic audience when writing cybersecurity content.

**Key Findings:**
Effective technical writing explains core concepts clearly before introducing detailed mechanics, always provides context, and assumes that readers are intelligent but not necessarily domain experts. Definitions and explanations of why a point matters are as important as the raw technical detail.

**Generative AI Contribution:**
ChatGPT provided a framework for judging whether content was too shallow or too dense and highlighted the need to connect technical sections back to project goals and evaluation criteria.

**Outcome:**
Sections with heavy jargon were revised to include definitions, context, and clearer explanations. A non-technical reader was asked to review sample sections to confirm that the main ideas were understandable without specialized background knowledge.

## Day 5 - 05 February 2026

**Activities Performed:**
Used Claude to clarify what a meaningful AI usage reflection should contain in an academic project report.

**Key Findings:**
A genuine reflection includes specific examples of prompts, honest discussion of strengths and limitations of AI tools, explanation of how outputs were verified, and description of what was learned. Generic statements like "AI was helpful" are insufficient.

**Generative AI Contribution:**
Claude distinguished between surface-level and deep reflection and proposed concrete elements that each team member should include in their AI reflection subsection.

**Outcome:**
Team members revised their AI reflection sections to add specific prompts, use cases, verification steps, and lessons learned. Each reflection was checked to ensure it moved beyond superficial comments.

## Day 6 - 06 February 2026

**Activities Performed:**
Consulted Gemini on citation styles (APA, MLA, IEEE) and which format is most appropriate for cybersecurity and threat intelligence reports.

**Key Findings:**
APA is commonly used for cybersecurity and IT security work because of its fit with technology and social science fields. Consistency across the document matters more than the specific style chosen, but proper formatting for each source type is essential.

**Generative AI Contribution:**
Gemini explained the characteristics of each citation style and provided APA examples for industry reports, websites, and academic papers.

**Outcome:**
The team adopted APA as the standard citation style. All references and in-text citations were updated accordingly and cross-checked with the Purdue OWL APA guide to verify correctness.

## Day 7 - 07 February 2026

**Activities Performed:**
Used Perplexity to understand how to write strong case studies in a cybersecurity threat report and how much detail to include about real-world incidents.

**Key Findings:**
Good case studies include a clear incident timeline, specific impact metrics, technical root causes, and explicit lessons learned. Simply repeating facts from news or reports does not count as analysis.

**Generative AI Contribution:**
Perplexity outlined these elements and illustrated how incidents such as the MOVEit breach or MGM attack could be structured to maximize analytical value.

**Outcome:**
Existing case studies in the report were revised to incorporate impact numbers and clearly stated lessons learned. Details were cross-checked against original breach sources to ensure factual accuracy.

## Day 8 - 08 February 2026

**Activities Performed:**
Consulted Claude on how to evaluate whether mitigation recommendations are realistic and practical rather than purely theoretical.

**Key Findings:**
Practical recommendations must consider cost, implementation difficulty, organizational context, and resource constraints. They

should be prioritized and justified, rather than presented as an unranked list of best practices.

**Generative AI Contribution:**
Claude provided criteria for assessing feasibility and priority, helping distinguish recommendations that could be implemented in a 3-week academic project from those more suitable for long-term enterprise programs.

**Outcome:**
The mitigation section was updated so that each recommendation included a priority level and clear justification. This made the section more actionable and aligned with realistic constraints.

## Day 9 - 09 February 2026

**Activities Performed:**
Used ChatGPT to create a final submission checklist for the cybersecurity threat intelligence report.

**Key Findings:**
Beyond core content, evaluators also look for formatting consistency, citation accuracy, visual quality, technical correctness, reflection components, page numbering, table of contents, and proper file naming.

**Generative AI Contribution:**
ChatGPT delivered a comprehensive checklist covering both content and presentation requirements, including several small but important items that could easily be overlooked.

**Outcome:**
The checklist was used for a systematic final review. It helped catch minor issues that might otherwise have reduced the overall grade, such as missing page numbers or inconsistent figure labels.

## Day 10 - 10 February 2026

**Activities Performed:**
Asked ChatGPT about common mistakes students make in cybersecurity project reports that cause them to lose marks.

**Key Findings:**
Frequent issues include unsupported claims without citations, inconsistent terminology, abrupt transitions, low-quality visuals, generic conclusions that do not tie back to analysis, and unverified or poorly integrated AI-generated content.

**Generative AI Contribution:**
ChatGPT provided a concise list of these pitfalls and explained why each one negatively affects evaluation.

**Outcome:**
A "mistakes to avoid" checklist was created from this guidance. The report was reviewed against it, and instances of these issues were identified and corrected before submission.

## Day 11 - 11 February 2026

**Activities Performed:**
Reflected on which AI prompt was most effective overall for project success.

**Key Findings:**
The initial prompt that generated the requirement checklist proved to be the most impactful, because it shaped how all later work was planned, reviewed, and verified against the assignment criteria.

**Generative AI Contribution:**
By acting as a virtual assessment coordinator, AI provided an evaluation framework that was reused throughout the project, rather than only helping with isolated sections.

**Outcome:**
This experience reinforced the value of using AI early to clarify expectations and create structured tools (like checklists) that guide the entire workflow.

## Day 12 - 12 February 2026

**Activities Performed:**
Reviewed how AI usage reflection skills evolved over the project, particularly around Prompt 5.

**Key Findings:**
Initial reflections tended to be superficial, but repeated prompting and clarification led to a much clearer understanding of what genuine AI reflection requires.

**Generative AI Contribution:**
AI feedback helped differentiate between merely mentioning tools and actually examining how, why, and with what limitations they were used.

**Outcome:**
The final AI reflection sections were more specific, honest, and analytically useful, better meeting academic expectations.

## Day 13 - 13 February 2026

**Activities Performed:**
Summarised lessons learned about what worked well when using AI as an assessment coordinator.

**Key Findings:**
Asking AI to explain evaluation criteria and reviewer expectations shifted the perspective from just finishing an assignment to thinking like an evaluator. This led to earlier detection of gaps and higher overall quality.

**Generative AI Contribution:**
AI helped design checklists, frameworks, and quality standards, enabling more precise feedback to teammates and a more structured review process.

**Outcome:**
The team benefitted from clearer guidance, more targeted revisions, and a better understanding of how high-quality cybersecurity reports are evaluated.

## Day 14 - 14 February 2026

**Activities Performed:**
Reflected on limitations and challenges encountered when using multiple AI tools.

**Key Findings:**
Some AI suggestions were too strict or more suited to professional reports than a 3-week academic project, and different tools sometimes offered conflicting advice. AI also could not directly judge whether specific content was good enough; human judgment was still required.

**Generative AI Contribution:**
AI provided rich criteria and examples but made it clear that final decisions had to be grounded in the actual assignment guidelines and context.

**Outcome:**
The project team learned to treat AI as a powerful support tool for frameworks and quality checks, while relying on their own judgment and reference materials to make final evaluation decisions.

# 3. Summary Statistics

| AI Tool | Total Prompts | Excellent | Good | Needed Work |
|---|---|---|---|---|
| ChatGPT | 4 | 3 | 1 | 0 |
| Perplexity | 2 | 2 | 0 | 0 |
| Claude | 3 | 2 | 1 | 0 |
| Gemini | 1 | 1 | 0 | 0 |

Table 1: AI tool usage breakdown across the 14-day project period

# 4. Conclusion

Over the 14-day period, AI tools served as valuable assessment and quality assurance resources. The most impactful use was creating evaluation frameworks early in the project, which shaped all subsequent work. AI helped translate assignment requirements into actionable checklists, identify quality standards for different report sections, and catch common mistakes before submission.

The experience demonstrated that AI is most effective when used to clarify expectations and build structured review processes, rather than simply generating content. Final judgment and contextual

decision-making remained essential human responsibilities throughout the project.